# Solving Capacitated Single Allocation p-Hub Location Problem using Metaheuristics Approaches

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## Abstract

This study investigates two metaheuristic approaches namely Tabu Search (TS) and Simulated Annealing (SA) applied to CSApHLP, a combinatorial optimisation problem arising in logistics and transportation network design. Both algorithms are evaluated across six benchmark datasets ranging from 10 to 100 nodes. Simulated Annealing employs a geometric cooling schedule with an adaptively computed initial temperature, while Tabu Search uses a tabu list with aspiration criteria to guide the search process. Results show that TS performs more consistently on smaller datasets, whereas SA achieves comparable or better solution quality on larger problems at up to $17\times$ lower computational time. Sensitivity analysis confirms that the inter-hub discount factor $\alpha$ has a predictable influence on total network cost across all instances.


## 1. Introduction

Many real-world decision problems in logistics, telecommunications, and transportation can be formulated as combinatorial optimisation problems (COPs), where the objective is to select the best solution from a finite but exponentially large set of possibilities. Such problems are characterised by two key requirements: (1) feasibility, meaning that all hard constraints must be satisfied, and (2) optimality, meaning that an objective function must be minimised or maximised over the feasible solution space (Hartmanis, 1982). A large class of COPs is NP-hard, implying that no polynomial-time exact algorithm is known for their general form, and exact approaches such as integer programming quickly become impractical as problem size grows.

This computational difficulty motivates the use of approximation techniques. Classical local search methods begin with an initial feasible solution and iteratively explore neighbouring solutions generated through defined transition operators. Improvements are accepted until no better neighbour can be identified (Lourenço et al., 2003). However, a key limitation of basic local search is its tendency to become trapped in local optima, with the final solution quality often depending heavily on the starting point (Reeves, 1993). Metaheuristic algorithms were developed to address this limitation. A metaheuristic provides a higher-level search strategy that guides the exploration process beyond local optimality, often allowing non-improving moves under controlled conditions in order to encourage broader exploration of the solution space (Glover, 1989).

An important application area for COP is the hub location problem. Hub-and-spoke network structures allows flows through intermediate hub facilities in order to exploit economies of scale on inter-hub transportation links (O'kelly, 1987b). Such networks are widely used in airline scheduling, freight transportation, and telecommunications systems (Klincewicz, 1998). The Capacitated Single Allocation $p$-Hub Location Problem (CSApHLP) extends this framework by fixing the number of hubs in advance and imposing capacity constraints.

### 1.1. Defining CSApHLP Constraints

The objective minimises total network transport and hub setup costs:

$$\min \underbrace{\sum_{i\in N}\sum_{j\in N} w_{ij}\left(c_{i,\phi(i)} + \alpha\, c_{\phi(i),\phi(j)} + c_{\phi(j),j}\right)}_{\text{network cost}} + \underbrace{\sum_{h\in N}\sum_{k\in K} F_k\, y_{hk}}_{\text{hub setup cost}},$$

where $\phi(i)$ is the hub assigned to node $i$, $\alpha \in (0, 1]$ is the inter-hub discount, and $F_k$ is the fixed cost of capacity level $k$. The complete variable definitions, constraint set, and capacity feasibility conditions are provided in Appendix A.1.

## 2. Related Work

**Tabu Search (TS)** was introduced by Glover (1989) as a memory-based extension of local search designed to avoid cycling and enable the exploration of new regions of the solution space. The algorithm maintains a tabu list that stores recently executed moves, temporarily prohibiting their reversal and thus encouraging the search to move beyond previously visited solutions. An aspiration criterion allows tabu restrictions to be overridden when a move produces a solution better than the current best, ensuring that promising

solutions are not discarded. In the context of hub location problems, Klincewicz (1998) demonstrated that TS can effectively explore structured neighbourhoods and produce high-quality solutions even for larger problem instances where exact methods become computationally infeasible.

**Simulated Annealing (SA)** is a stochastic local search technique that allows non-improving moves to escape local optima (Bertsimas & Tsitsiklis, 1993). The search begins with an initial temperature $T_0$, which controls the level of exploration during the early stages of the algorithm and helps prevent premature convergence. At each iteration, a neighbouring solution $\mathbf{s}'$ is sampled. If $\Delta f = f(\mathbf{s}') - f(\mathbf{s}) < 0$, the move is accepted; otherwise it is accepted with Metropolis probability $\exp(-\Delta f/T)$ (Abdinnour-Helm, 2001). The choice of cooling schedule plays a critical role in the performance of SA. Abdinnour-Helm (2001) employed a geometric cooling schedule for the $p$-hub median problem, demonstrating that a gradually decreasing temperature enables the algorithm to generate competitive hub configurations across different instance sizes. Similarly, Azizi & Zolfaghari (2004) emphasised the importance of cooling schedule design by proposing an adaptive trajectory-based approach, showing that non-monotonic temperature schedules can outperform traditional decreasing schemes.

# 3. Methodology

## 3.1. Experimentation Setup

The problem instances constructed by varying the number of hubs (p) and inter-hub discount factor ($\alpha$), yields 22 distinct configurations. All experiments are conducted on the benchmark instances summarised in Appendix table 3, spanning two real-world datasets (CAB O'kelly (1987a), TR Tan & Kara (2007)) and one randomly generated problem (RGP100), with solutions encoded as integer assignment vectors using the closest-hub approach (based on lowest transport cost). For a detailed explanation of the *dataset description* and *solution representation* refer Appendix A.2 and A.3.

### 3.1.1. Neighborhood Selection

Two neighborhood structures are used during search:

**Hub replacement move ($\mathcal{N}_1$):** A spoke assigned to a hub is promoted to replace that hub, and nodes previously assigned to the old hub are redirected accordingly.

**Spoke reassignment move ($\mathcal{N}_2$):** A non-hub node is reassigned from its current hub to a different open hub.

## 3.2. Tabu Search Algorithm

At each iteration, the full neighbourhood $\mathcal{N}(\mathbf{s}) = \mathcal{N}_1(\mathbf{s}) \cup \mathcal{N}_2(\mathbf{s})$ is generated and the best admissible neighbour $\mathbf{s}'$ is selected subject to tabu restrictions. A move is tabu if it reverses a recently performed transition recorded in the tabu list $\mathcal{T}$; this is overridden by the aspiration criterion when $f(\mathbf{s}') < f^\star$, ensuring globally improving moves are never rejected. The current solution is updated unconditionally to $\mathbf{s}'$, and the best solution $\mathbf{s}^\star$ is updated whenever an improvement is found.

### 3.2.1. Stopping Criteria

Termination is governed by a maximum iteration count $N$ (N-rule) and a no-improvement window $W$ (W-rule), following Corominas (2023). The value of $N$ is scaled with problem size, and $W$ is set to $10\%$ of $N$.

## 3.3. Simulated Algorithm

At each iteration, a single neighbour $\mathbf{s}'$ is sampled uniformly at random from $\mathcal{N}(\mathbf{s}) = \mathcal{N}_1(\mathbf{s}) \cup \mathcal{N}_2(\mathbf{s})$. If $\Delta f = f(\mathbf{s}') - f(\mathbf{s}) < 0$, the move is accepted outright; otherwise it is accepted with a probability $\exp(-\Delta f/T)$. The best solution $\mathbf{s}^\star$ is updated whenever an accepted move yields an improvement. The temperature is reduced geometrically after every iteration via $T_{k+1} = \beta T_k$.

### 3.3.1. Initial Temperature and Cooling Schedule

Following Abdinnour-Helm (2001), the initial temperature is set as:

$$T_0 = -\frac{\overline{\Delta f}}{\ln(p_0)}, \tag{1}$$

where $\overline{\Delta f}$ is the mean cost increase over a sample of random neighbours, ensuring an average uphill move is accepted with probability $p_0 = 0.8$ at the start of the search.

A geometric cooling schedule is applied after every iteration, where $\beta$ is the cooling rate (Abdinnour-Helm, 2001; Laarhoven & Aarts, 1987):

$$T_{k+1} = \beta T_k, \quad 0 < \beta < 1 \tag{2}$$

### 3.3.2. Stopping Criteria

The maximum iteration count $N$ is the primary stopping criterion. The final temperature $T_f$ acts as a numerical safeguard, ensuring the temperature remains strictly positive throughout the search (Abdinnour-Helm, 2001).

## 3.4. Parameter Tuning For TS and SA

In this study, tabu tenure for TS and $\beta$ for SA is selected empirically from a set of candidate values. These candidate values are evaluated over five independent runs, and the value yielding the best objective value is selected.
**Parameter Selection:** see table 6 and 7 in Appendix A.

*Table 1.* Computational results — Tabu Search (TS). Costs reported in millions (M).

| **Problem**[(1)] | **p**[(2)] | $\alpha^{(3)}$ | **Solution Config./Location of Hubs** | **TNC**[(4)] **(M)** | **Avg. TNC**[(5)] **(M)** | **Time**[(6)] **(sec)/ # of Iter. (for best cost)** |
|---|---|---|---|---|---|---|
| CAB10 | 3 | 0.3 | [4,3,3,4,4,4,7,4,4,7]<br>3(L1); 4(L3); 7(L1) | 811.77 | 816.85 | 0.068 / 25 |
| | | 0.7 | [4,2,2,4,4,4,7,4,4,7]<br>2(L1); 4(L3); 7(L1) | 966.62 | 973.36 | 0.079 / 23 |
| | 4 | 0.3 | [4,6,6,4,4,6,7,8,6,7]<br>4(L2); 6(L2); 7(L1); 8(L1) | 758.87 | 770.95 | 0.090 / 26 |
| | | 0.7 | [5,6,6,4,5,6,7,4,6,7]<br>4(L2); 5(L1); 6(L2); 7(L1) | 937.02 | 953.56 | 0.076 / 26 |
| CAB20 | 3 | 0.3 | [4,17,17,4,4,4,4,4,4,4,4,12,4,17,4,4,17,17,12,17]<br>4(L3); 12(L1); 17(L3) | 4873.99 | 5328.46 | 0.392 / 61 |
| | | 0.7 | [1,17,17,4,4,4,4,4,4,1,4,4,1,1,4,1,17,17,4,17]<br>1(L1); 4(L3); 17(L2) | 6385.32 | 6549.23 | 0.382 / 66 |
| | 5 | 0.3 | [4,17,17,4,4,4,7,7,4,7,4,12,4,14,4,7,17,17,12,17]<br>4(L2); 7(L1); 12(L1); 14(L1); 17(L2) | 3512.66 | 3635.02 | 0.627 / 66 |
| | | 0.7 | [1,17,17,4,4,4,7,7,4,7,4,12,1,1,4,1,17,17,12,17]<br>1(L1); 4(L2); 7(L1); 12(L1); 17(L2) | 5390.07 | 5559.68 | 0.648 / 69 |
| CAB25 | 3 | 0.3 | [4,18,18,4,4,18,4,4,4,4,4,12,4,18,4,4,18,18,12,<br>18,4,12,12,18,18]<br>4(L2); 12(L1); 18(L3) | 7454.18 | 7739.95 | 0.922 / 116 |
| | | 0.7 | [2,2,2,4,4,2,4,4,4,4,4,12,4,2,4,4,2,2,12,2,4,12,<br>12,2,2]<br>2(L3); 4(L2); 12(L1) | 9661.54 | 10315.08 | 0.830 / 104 |
| | 5 | 0.3 | [4,17,17,4,4,4,7,7,4,7,4,12,4,14,4,7,17,17,12,17,<br>4,12,12,14,17]<br>4(L2); 7(L1); 12(L1); 14(L1); 17(L2) | 5672.47 | 5987.56 | 1.354 / 94 |
| | | 0.7 | [1,18,18,4,4,4,7,7,4,7,4,12,1,1,4,1,18,18,12,18,<br>4,12,12,1,18]<br>1(L1); 4(L2); 7(L1); 12(L1); 18(L2) | 8529.13 | 9007.97 | 1.317 / 102 |
| TR40 | 4 | 0.3 | 3(L2); 14(L2); 19(L1); 22(L1) | 9335.24 | 9745.27 | 2.311 / 106 |
| | | 0.7 | 1(L2); 3(L2); 22(L1); 24(L1) | 12004.21 | 13033.70 | 2.349 / 113 |
| | 6 | 0.3 | 1(L1); 3(L2); 5(L1); 7(L1); 24(L2); 31(L1) | 8654.28 | 9337.80 | 3.453 / 128 |
| | | 0.7 | 1(L2); 3(L2); 4(L1); 12(L1); 24(L1); 29(L1) | 11373.24 | 11789.02 | 3.531 / 114 |
| TR55 | 5 | 0.3 | 1(L1); 4(L2); 15(L1); 26(L2); 33(L1) | 20553.95 | 21776.43 | 5.304 / 125 |
| | | 0.7 | 1(L1); 4(L2); 14(L1); 17(L1); 30(L2) | 31129.20 | 32110.52 | 4.816 / 131 |
| | 7 | 0.3 | 4(L1); 15(L1); 26(L2); 31(L1); 33(L1);<br>37(L1); 55(L1) | 18133.74 | 20672.55 | 6.874 / 134 |
| | | 0.7 | 1(L1); 4(L1); 5(L1); 19(L1); 30(L2); 33(L1);<br>37(L1) | 28556.85 | 29373.45 | 6.527 / 121 |
| RGP100 | 9 | 0.3 | 4(L1); 6(L1); 10(L1); 11(L1); 15(L1); 20(L1);<br>52(L1); 60(L1); 69(L1) | 126224.72 | 127627.15 | 27.715 / 146 |
| | 12 | 0.3 | 9(L1); 10(L1); 28(L1); 40(L1); 55(L1);<br>61(L1); 63(L1); 74(L1); 80(L1); 86(L1);<br>90(L1); 94(L1) | 122407.79 | 123058.06 | 36.635 / 161 |

[(1)]Number of nodes/dataset; [(2)]Number of hubs; [(3)]Discount factor [(4)]Total Network Cost;

[(5)]Average network cost over 5 runs; [(6)]Computational time/number of iterations.

*Table 2.* Computational results — Simulated Annealing (SA).

| **Problem**[(1)] | **p**[(2)] | $\alpha^{(3)}$ | **Solution Config./Location of Hubs** | **TNC**[(4)] **(M)** | **Avg. TNC**[(5)] **(M)** | **Time**[(6)] **(sec)/ # of Iter. (for best cost)** |
|---|---|---|---|---|---|---|
| CAB10 | 3 | 0.3 | [4,3,3,4,4,4,7,4,4,7]<br>3(L1); 4(L3); 7(L1) | 811.77 | 832.63 | 0.143 / 1000 |
| | | 0.7 | [4,4,6,4,4,6,7,4,4,7]<br>4(L3); 6(L1); 7(L1) | 979.28 | 1005.00 | 0.140 / 1000 |
| | 4 | 0.3 | [6,6,6,4,6,6,7,8,6,7]<br>4(L2); 6(L3); 7(L1); 8(L1) | 801.34 | 827.73 | 0.143 / 1000 |
| | | 0.7 | [1,6,6,6,6,6,7,8,6,7]<br>1(L1); 6(L3); 7(L1); 8(L1) | 965.53 | 992.94 | 0.140 / 1000 |
| CAB20 | 3 | 0.3 | [4,17,17,4,4,4,4,4,4,4,4,12,4,17,4,4,17,17,12,17]<br>4(L3); 12(L1); 17(L3) | 4873.99 | 4988.71 | 0.287 / 2000 |
| | | 0.7 | [5,17,17,5,5,5,7,7,5,7,7,7,5,5,5,7,17,17,7,17]<br>5(L2); 7(L2); 17(L2) | 6375.75 | 6452.06 | 0.291 / 2000 |
| | 5 | 0.3 | [1,17,17,4,6,6,1,4,6,1,4,12,4,1,4,1,17,17,12,6]<br>1(L1); 4(L2); 6(L1); 12(L1); 17(L2) | 3818.56 | 3987.09 | 0.292 / 2000 |
| | | 0.7 | [1,17,17,4,6,6,1,4,6,1,4,12,4,1,4,1,17,17,12,6]<br>1(L1); 4(L2); 6(L1); 12(L1); 17(L2) | 5441.83 | 5572.37 | 0.286 / 2000 |
| CAB25 | 3 | 0.3 | [4,18,18,4,4,18,4,4,4,4,4,12,4,18,4,4,18,18,12,<br>18,4,12,12,18,18]<br>4(L2); 12(L1); 18(L3) | 7454.18 | 7592.75 | 0.359 / 2500 |
| | | 0.7 | [2,2,2,4,4,2,4,4,4,4,4,12,4,2,4,4,2,2,12,2,4,12,<br>12,2,2]<br>2(L3); 4(L2); 12(L1) | 9661.54 | 9870.24 | 0.368 / 2500 |
| | 5 | 0.3 | [24,17,17,4,4,4,7,7,4,7,4,12,7,24,4,7,17,17,12,17,<br>4,12,12,24,17]<br>4(L2); 7(L1); 12(L1); 17(L2); 24(L1) | 5694.51 | 6004.27 | 0.371 / 2500 |
| | | 0.7 | [13,18,18,4,4,18,13,13,4,13,13,12,13,14,4,13,18,<br>18,12,18,13,12,12,14,18]<br>4(L1); 12(L1); 13(L1); 14(L1); 18(L2) | 8597.04 | 8673.19 | 0.379 / 2500 |
| TR40 | 4 | 0.3 | 1(L2); 3(L2); 24(L2); 35(L1) | 9246.25 | 9656.58 | 0.630 / 4000 |
| | | 0.7 | 1(L2); 3(L2); 22(L1); 24(L1) | 12004.21 | 12525.64 | 0.617 / 4000 |
| | 6 | 0.3 | 1(L1); 3(L1); 19(L1); 22(L1); 31(L1); 32(L1) | 7802.76 | 8029.40 | 0.633 / 4000 |
| | | 0.7 | 1(L2); 3(L2); 4(L1); 20(L1); 22(L1); 24(L1) | 11343.85 | 11836.07 | 0.652 / 4000 |
| TR55 | 5 | 0.3 | 1(L1); 4(L2); 15(L1); 26(L2); 33(L1) | 20553.95 | 21124.94 | 0.967 / 5500 |
| | | 0.7 | 1(L2); 4(L2); 5(L1); 30(L2); 33(L1) | 30588.47 | 30950.12 | 0.983 / 5500 |
| | 7 | 0.3 | 1(L1); 4(L1); 14(L1); 15(L1); 26(L2); 31(L1);<br>33(L1) | 18217.85 | 18678.61 | 0.982 / 5500 |
| | | 0.7 | 1(L1); 4(L1); 15(L1); 30(L2); 31(L1); 33(L1);<br>37(L1) | 28041.81 | 28926.63 | 1.012 / 5500 |
| RGP100 | 9 | 0.3 | 4(L1); 22(L1); 26(L1); 34(L1); 38(L1);<br>46(L1); 58(L1); 63(L1); 86(L1) | 125692.60 | 126318.29 | 2.368 / 10000 |
| | 12 | 0.3 | 4(L1); 5(L1); 7(L1); 13(L1); 15(L1); 27(L1);<br>38(L1); 41(L1); 72(L1); 77(L1); 88(L1);<br>97(L1) | 120395.71 | 122053.85 | 2.382 / 10000 |

[(1)]Number of nodes/dataset; [(2)]Number of hubs; [(3)]Discount factor [(4)]Total Network Cost;

[(5)]Average network cost over 5 runs; [(6)]Computational time/number of iterations.

# 4. Results and Discussion

## 4.1. Results for TS

Table 1 reports the best and average Total Network Cost (TNC) obtained by Tabu Search (TS) across all benchmark instances. TS consistently produces high-quality solutions, with best costs ranging from 811.77M on CAB10 (p=3) to 126,224.72M on RGP100 (p=9). For smaller instances (CAB10–CAB25), the average costs remain close to the best values, indicating stable search behaviour and low variance across runs (see Table 6 in Appendix A).

For larger number of nodes such as TR55 and RGP100, the gap between best and average costs becomes more noticeable. For example, TR55 (p=7) yields a best cost of 18,133.74M in contrary to an average of 20,672.55M. This suggests increased sensitivity to initial solutions as the size of the solution space grows. In terms of computational performance, TS terminates within a reasonable number of iterations across all instances, with runtimes ranging from $< 0.1$ seconds for CAB10 to $\approx 37$ seconds for RGP100 (p=12).

## 4.2. Results for SA

Table 2 presents the results obtained using Simulated Annealing (SA) for the same benchmarks. SA achieves best costs in comparison to TS on most instances, and matches TS (Tie) in several cases such as CAB10 (p=3, $\alpha = 0.3$) and CAB25 (p=3).

Figure 1 illustrates the cooling curves for CAB10 under $\alpha = 0.3$. Higher $\beta$ values maintain elevated temperatures for more iterations, allowing greater exploration before convergence, whereas lower $\beta$ values result in faster cooling. The chosen $\beta$ parameters therefore provide a gradual and controlled reduction in temperature, allowing the algorithm sufficient opportunity to explore the neighbourhood before converging. This behaviour is consistent with the theoretical motivation for geometric cooling schedules as proposed by Abdinnour-Helm (2001).

On larger number of node size, SA's average costs are notably competitive. For example, on RGP100 (p=12), SA achieves an average cost of 122,053.85M compared with TS's 123,058.06M, while requiring significantly less runtime (approximately 2.4 seconds versus 36.6 seconds).

## 4.3. Discussion

### 4.3.1. Head-To-Head Comparison For TS and SA

Figure 2 summarises the per-instance cost differences between TS and SA. Overall the algorithms are evenly matched with TS outperforming SA in 8 instances, SA outperforming TS in 8, and 6 instances resulting in ties.

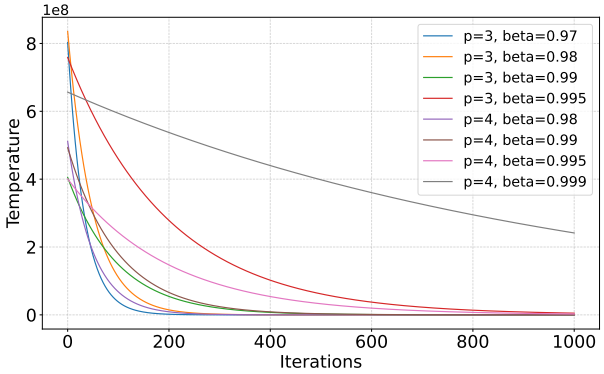


*Figure 1.* SA Cooling curve for CAB10 dataset, $\alpha = 0.3$

However, a clear trend emerges with respect to instance size (# of nodes in the dataset). TS performs more consistently on smaller datasets (CAB10–CAB25), where its deterministic best-admissible move selection efficiently exploits the relatively compact neighbourhood structure.

In contrast, SA performs better on larger datasets (TR40, TR55, and RGP100). As its probabilistic acceptance mechanism allows broader exploration of the solution space, thereby helping the algorithm avoid premature convergence that may occur in TS when operating with a fixed tabu tenure.

The computational efficiency advantage of SA becomes particularly evident for larger datasets. On RGP100, SA achieves equal or better solution quality while requiring up to $17\times$ less runtime. This is practically significant given the NP-hard nature of the hub location problem and is consistent with findings by Ghaffarinasab et al. (2018), who reported that SA can produce competitive hub configurations even when neighbourhood evaluation becomes computationally expensive.

### 4.3.2. Sensitivity Analysis For Discount Factor

Figure 3 examines how objective costs change as the discount factor $(\alpha)$ increases from 0.3 to 0.7. Across all datasets and both algorithms, costs rise consistently with increasing $\alpha$, which confirms its direct influence on the inter-hub routing component of the objective function.

The magnitude of this cost increase depends on both the node size and the number of hubs for the problem. For example, within CAB20 the cost increase for p=5 (53.4% under TS) is substantially larger than for p=3 (30.2%). This occurs because a larger number of hubs creates more inter-hub connections, which benefit more strongly from the lower discount at $\alpha = 0.3$. TR55 (p=7) shows the highest overall sensitivity (58.4% under SA), reflecting the denser flow structure of the TR network. Importantly, both algorithms

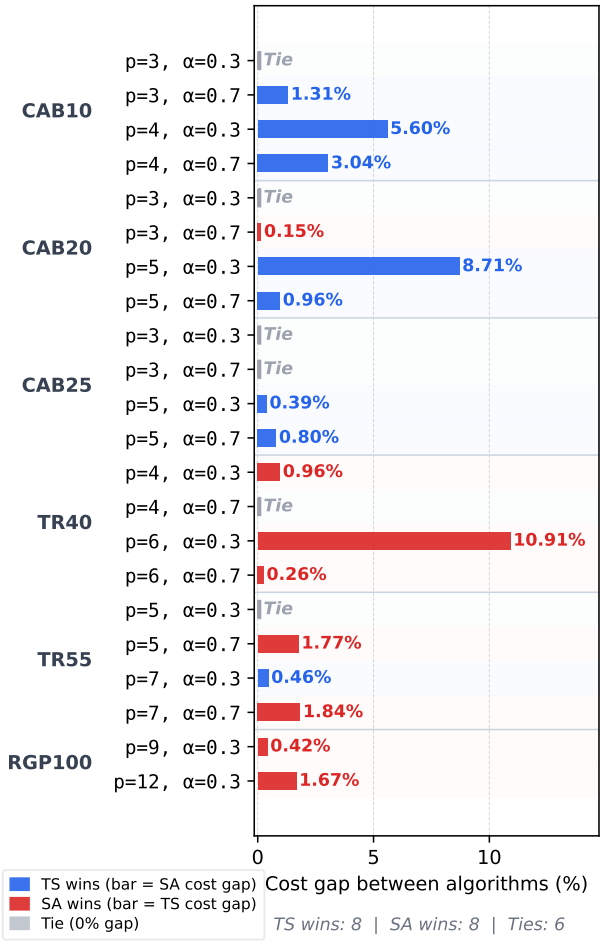


*Figure 2.* TS vs. SA — per instance cost gap

shows nearly identical responses to changes in $\alpha$, which indicates that this behaviour arises from the underlying cost structure of the problem rather than from algorithm-specific effects.

## 5. Conclusion

This study applied Tabu Search and Simulated Annealing to the CSApHLP across six benchmark datasets of varying scale. Both algorithms produce competitive solutions, with TS performing more reliably on smaller datasets and SA demonstrating stronger scalability on larger problems. In particular, SA achieves comparable or better solution quality while requiring up to $17\times$ lower runtime on the RGP100 dataset. Sensitivity analysis further confirms that the inter-hub discount factor $\alpha$ has a consistent and predictable influence on total network cost through its direct effect on the inter-hub transportation component of the objective function, independent of the algorithm used. Overall, the results demonstrate that metaheuristic approaches provide a practical and effective alternative to exact optimisation methods for this class of NP-hard hub location problems.

**Limitations and Future Work:** Parameter tuning was conducted empirically using a limited set of candidate values, and algorithm performance on datasets larger than RGP100 remains unexplored. Future work could investigate adaptive parameter control strategies, hybrid metaheuristic approaches, or benchmarking against exact methods on larger standardised datasets.

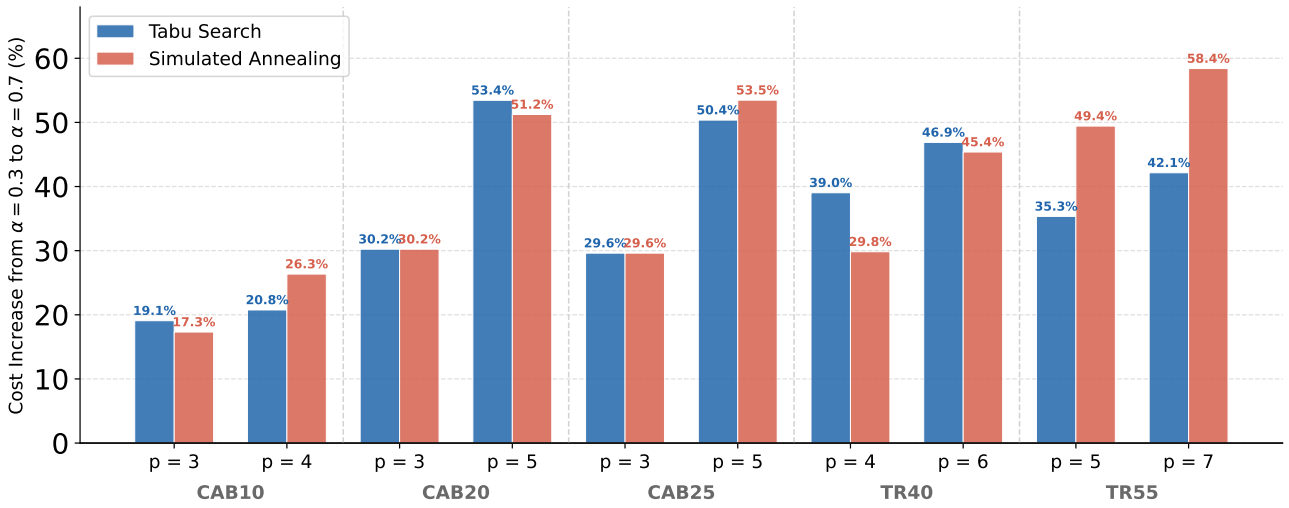


*Figure 3.* Sensitivity analysis for discount factor $(\alpha)$. Percentage cost increase from $\alpha = 0.3$ to $\alpha = 0.7$ across benchmark datasets and algorithms. E.g. the percentage cost increase for TS CAB10 (p=3) is 19.1% as $Cost(\alpha = 0.3) = 811.77M$ and $Cost(\alpha = 0.7) = 966.62M$

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# A. APPENDIX

## A.1. CSApHLP: Full Formulation

Let $N = \{1, \ldots, n\}$ be nodes, $p$ the predetermined number of hubs, and $K = \{1, 2, 3\}$ the capacity levels (small, medium, large). For any two nodes $i$ and $j$, $w_{ij}$ and $c_{ij}$ represent the demand/flow and transport cost respectively, and $\alpha \in (0, 1]$ is the inter-hub discount.

**Binary decision variables:**

$$x_{ih} = \begin{cases} 1, & \text{node } i \text{ is allocated to hub } h, \\ 0, & \text{otherwise,} \end{cases}$$

$$z_h = x_{hh},$$

$$y_{hk} = \begin{cases} 1, & \text{hub } h \text{ uses capacity level } k, \\ 0, & \text{otherwise.} \end{cases}$$

**Core constraints:**

$$\sum_{h \in N} x_{ih} = 1 \quad \forall i \in N$$

(single allocation),

$$x_{ih} \leq z_h \quad \forall i, h \in N$$

(allocate only to opened hubs),

$$\sum_{h \in N} z_h = p$$

(exactly $p$ hubs opened),

$$\sum_{k \in K} y_{hk} = z_h \quad \forall h \in N$$

(one capacity level per opened hub).

Using the assignment rule in the implementation, hub load is:

$$L_h = \sum_{i \in N} x_{ih} \left( \sum_{j \in N} w_{ji} \right).$$

Hub load is computed solely from the aggregate incoming flow of nodes directly allocated to hub $h$; transit flows arising from inter-hub routing are not considered.

**Capacity feasibility:**

$$L_h \leq \sum_{k \in K} U_k \, y_{hk} \quad \forall h \in N,$$

where $U_k$ is the capacity threshold of level $k$.

## A.2. Dataset Description

The benchmark instances used in this study are derived from two real-world datasets, namely CAB O'kelly (1987a) and TR Tan & Kara (2007), as well as a randomly generated problem (RGP).

For each dataset, multiple problem instances are generated by varying the number of hubs to be opened $p$ and the inter-hub discount factors $\alpha$. The considered problem instances are summarised in Table 3.

*Table 3.* Hub configurations and discount factors for benchmark datasets

| Dataset | Number of Hubs ($p$) | Discount Factor ($\alpha$) |
|---|---|---|
| CAB10 | 3, 4 | 0.3, 0.7 |
| CAB20 | 3, 5 | 0.3, 0.7 |
| CAB25 | 3, 5 | 0.3, 0.7 |
| TR40 | 4, 6 | 0.3, 0.7 |
| TR55 | 5, 7 | 0.3, 0.7 |
| RGP100 | 9, 12 | 0.3 |

## A.3. Solution Representation

A solution is encoded as an integer vector such that a node is a hub iff it is self-assigned, and every non-hub appears exactly once with one hub label (single allocation) in the vector. In cost evaluation, each opened hub is mapped to the smallest feasible level, namely small (L1), medium (L2), and large (L3), with a subsequent fixed setup cost. The fixed costs for each capacity level are summarised in table 4, while the dataset-specific capacity values are provided in table 5.

*Table 4.* Fixed setup costs for hub capacity levels

| Capacity Level | Fixed Cost |
|---|---|
| L1 (Small) | 50,000,000 |
| L2 (Medium) | 100,000,000 |
| L3 (Large) | 150,000,000 |

*Table 5.* Dataset-wise hub capacity values

| Dataset | L1 (Small) | L2 (Medium) | L3 (Large) |
|---|---|---|---|
| CAB10 | 199,805.2 | 399,610.4 | 849,172.1 |
| CAB20 | 1,150,918.8 | 2,301,837.6 | 4,891,404.9 |
| CAB25 | 1,708,001.2 | 3,416,002.4 | 7,259,005.1 |
| TR40 | 3,157,851.46 | 6,315,702.92 | 13,420,868.71 |
| TR55 | 8,150,485.4 | 16,300,970.8 | 34,639,562.95 |
| RGP100 | 19,800,492.8 | 39,600,985.6 | 84,152,094.4 |

*Table 6.* Tabu Search parameter tuning results ($\alpha = 0.3$). Best configuration per instance shown in **bold**.

| Dataset | p | Tabu Tenure | Best Cost (in M) | Avg Cost (in M) | Std Dev (in M) | Avg Time (in s) | # of Iter. (for best cost) |
|---|---|---|---|---|---|---|---|
| CAB10 | 3 | 4 | 811.77 | 866.28 | 59.04 | 0.072 | 25 |
| | | **5** | **811.77** | **821.93** | **12.43** | **0.073** | **23** |
| | | 6 | 811.77 | 867.24 | 58.19 | 0.175 | 34 |
| | | 7 | 811.77 | 865.56 | 56.84 | 0.137 | 46 |
| CAB10 | 4 | 6 | 763.95 | 785.13 | 15.00 | 0.088 | 21 |
| | | 7 | 758.87 | 805.68 | 66.94 | 0.091 | 34 |
| | | 8 | 764.44 | 781.48 | 24.43 | 0.073 | 24 |
| | | **9** | **758.87** | **772.73** | **9.21** | **0.109** | **42** |
| CAB20 | 3 | 4 | 5267.56 | 5471.91 | 269.92 | 0.536 | 69 |
| | | 5 | 4873.99 | 5234.96 | 412.34 | 0.442 | 55 |
| | | **6** | **4873.99** | **5146.14** | **144.21** | **0.459** | **56** |
| | | 7 | 4873.99 | 5298.78 | 256.83 | 0.550 | 60 |
| CAB20 | 5 | **8** | **3512.66** | **3810.89** | **263.71** | **0.722** | **61** |
| | | 9 | 3512.66 | 4077.18 | 463.04 | 0.788 | 71 |
| | | 10 | 3512.66 | 3976.09 | 426.51 | 0.656 | 71 |
| | | 11 | 3512.66 | 3924.59 | 438.67 | 0.745 | 87 |
| CAB25 | 3 | 5 | 7454.18 | 8293.94 | 807.34 | 0.909 | 94 |
| | | 6 | 7454.18 | 7988.51 | 526.56 | 0.869 | 92 |
| | | 7 | 7454.18 | 8097.69 | 796.30 | 0.881 | 87 |
| | | **8** | **7454.18** | **7454.18** | **0.00** | **0.938** | **95** |
| CAB25 | 5 | **8** | **5672.47** | **5759.29** | **173.65** | **1.412** | **97** |
| | | 9 | 5672.47 | 6013.47 | 417.94 | 1.369 | 88 |
| | | 10 | 5672.47 | 5932.44 | 223.91 | 1.520 | 96 |
| | | 11 | 5672.47 | 6170.93 | 449.63 | 1.515 | 105 |
| TR40 | 4 | 7 | 9246.25 | 9831.73 | 298.59 | 2.686 | 125 |
| | | **8** | **9246.25** | **9533.15** | **316.72** | **2.773** | **131** |
| | | 9 | 9261.55 | 9953.62 | 570.92 | 2.796 | 113 |
| | | 10 | 9246.25 | 9720.06 | 670.44 | 2.698 | 128 |
| TR40 | 6 | 10 | 7860.62 | 8278.76 | 340.20 | 4.279 | 117 |
| | | 11 | 7908.14 | 8419.27 | 301.33 | 4.236 | 115 |
| | | **12** | **7762.69** | **8319.41** | **340.93** | **4.233** | **125** |
| TR55 | 5 | 11 | 20553.95 | 22790.93 | 1177.79 | 5.276 | 118 |
| | | 12 | 22637.99 | 22720.63 | 110.45 | 5.337 | 125 |
| | | **13** | **20553.95** | **21685.86** | **1186.27** | **5.573** | **126** |
| TR55 | 7 | 12 | 20403.77 | 21641.09 | 1561.49 | 7.095 | 117 |
| | | 13 | 18984.73 | 20983.04 | 1747.47 | 6.909 | 122 |
| | | 14 | 18848.09 | 20060.09 | 900.06 | 7.361 | 132 |
| | | **15** | **17949.06** | **19797.84** | **960.84** | **7.640** | **116** |
| RGP100 | 9 | **17** | **125651.23** | **127021.02** | **1239.85** | **31.362** | **162** |
| | | 18 | 127659.10 | 128318.84 | 645.21 | 29.363 | 164 |
| | | 19 | 128432.78 | 129038.51 | 467.96 | 29.655 | 166 |
| | | 20 | 126916.79 | 127857.04 | 869.99 | 31.329 | 178 |
| RGP100 | 12 | 21 | 121330.43 | 122648.68 | 833.03 | 40.791 | 163 |
| | | 22 | 121926.18 | 122663.55 | 646.60 | 38.639 | 162 |
| | | 23 | 120815.13 | 122336.80 | 1159.75 | 41.773 | 218 |
| | | **24** | **120684.40** | **122223.47** | **897.03** | **41.529** | **161** |

*Table 7.* Simulated Annealing parameter tuning results ($\alpha = 0.3$). $p_0 = 0.8$ is taken as a fixed parameter and the initial temperature is computed adaptively. Best configuration per instance is shown in **bold**.

| Dataset | p | $\beta$ | Best Cost (in M) | Avg Cost (in M) | Std Dev (in M) | Avg Time (in s) | # of Iter. (for best cost) |
|---|---|---|---|---|---|---|---|
| CAB10 | 3 | 0.970 | 816.56 | 850.09 | 26.41 | 0.152 | 1000 |
| | | 0.980 | 811.77 | 860.98 | 38.29 | 0.150 | 1000 |
| | | **0.990** | **811.77** | **835.02** | **21.14** | **0.153** | **1000** |
| | | 0.995 | 811.77 | 842.16 | 24.73 | 0.150 | 1000 |
| CAB10 | 4 | 0.980 | 763.95 | 771.34 | 14.05 | 0.150 | 1000 |
| | | 0.990 | 764.44 | 786.27 | 26.91 | 0.147 | 1000 |
| | | 0.995 | 763.95 | 769.90 | 8.79 | 0.159 | 1000 |
| | | **0.999** | **758.87** | **812.08** | **35.85** | **0.181** | **1000** |
| CAB20 | 3 | 0.970 | 4873.99 | 5713.82 | 1252.22 | 0.240 | 2000 |
| | | 0.980 | 4873.99 | 5146.96 | 261.94 | 0.293 | 2000 |
| | | 0.990 | 4924.62 | 5107.79 | 167.63 | 0.291 | 2000 |
| | | **0.995** | **4873.99** | **4998.83** | **133.52** | **0.292** | **2000** |
| CAB20 | 5 | 0.960 | 3645.29 | 4015.12 | 358.65 | 0.292 | 2000 |
| | | **0.970** | **3512.66** | **3853.60** | **279.22** | **0.297** | **2000** |
| | | 0.980 | 3645.29 | 3949.79 | 190.25 | 0.293 | 2000 |
| | | 0.990 | 3818.56 | 3929.30 | 114.34 | 0.296 | 2000 |
| CAB25 | 3 | 0.750 | 7454.18 | 7989.77 | 765.54 | 0.356 | 2478 |
| | | 0.800 | 7454.18 | 7611.11 | 262.21 | 0.367 | 2500 |
| | | **0.850** | **7454.18** | **7456.50** | **4.63** | **0.373** | **2500** |
| | | 0.900 | 7454.18 | 7470.79 | 22.22 | 0.361 | 2500 |
| CAB25 | 5 | **0.960** | **5672.47** | **6158.19** | **319.99** | **0.373** | **2500** |
| | | 0.970 | 5672.47 | 6201.31 | 489.87 | 0.386 | 2500 |
| | | 0.980 | 5672.47 | 6311.45 | 360.24 | 0.373 | 2500 |
| | | 0.990 | 5694.51 | 6066.31 | 277.82 | 0.370 | 2500 |
| TR40 | 4 | 0.970 | 9261.55 | 9301.90 | 34.57 | 0.623 | 4000 |
| | | 0.980 | 9261.55 | 9530.09 | 452.38 | 0.623 | 4000 |
| | | **0.990** | **9246.25** | **9747.67** | **431.96** | **0.667** | **4000** |
| | | 0.995 | 9335.24 | 9676.18 | 281.38 | 0.646 | 4000 |
| TR40 | 6 | **0.980** | **7762.69** | **7941.83** | **330.78** | **0.680** | **4000** |
| | | 0.990 | 7787.56 | 8022.34 | 287.83 | 0.684 | 4000 |
| | | 0.995 | 7762.69 | 8010.48 | 258.44 | 0.720 | 4000 |
| | | 0.999 | 8355.12 | 9013.70 | 562.08 | 0.703 | 4000 |
| TR55 | 5 | 0.970 | 20553.95 | 21576.90 | 829.70 | 0.967 | 5500 |
| | | 0.980 | 20553.95 | 22307.57 | 1067.82 | 0.976 | 5500 |
| | | 0.990 | 20553.95 | 21905.88 | 916.75 | 0.977 | 5500 |
| | | **0.995** | **20553.95** | **21531.18** | **729.87** | **0.986** | **5500** |
| TR55 | 7 | 0.980 | 18241.67 | 19221.61 | 1027.50 | 0.983 | 5500 |
| | | 0.990 | 18333.75 | 19308.67 | 989.96 | 0.981 | 5500 |
| | | **0.995** | **17870.77** | **18801.24** | **1099.00** | **1.006** | **5500** |
| | | 0.999 | 18317.46 | 19265.37 | 622.20 | 0.984 | 5500 |
| RGP100 | 9 | 0.990 | 125136.60 | 126301.51 | 747.94 | 2.391 | 10000 |
| | | **0.995** | **124937.84** | **126039.30** | **592.40** | **2.412** | **10000** |
| | | 0.999 | 125024.88 | 125635.82 | 562.23 | 2.397 | 10000 |
| | | 0.9995 | 125367.03 | 127912.22 | 1290.87 | 2.380 | 10000 |
| RGP100 | 12 | **0.990** | **120150.03** | **120976.33** | **669.09** | **2.400** | **10000** |
| | | 0.995 | 121454.70 | 122323.04 | 851.76 | 2.393 | 10000 |
| | | 0.999 | 120448.24 | 121305.51 | 455.75 | 2.450 | 10000 |
| | | 0.9995 | 122333.93 | 123182.55 | 459.78 | 2.377 | 10000 |